Ask BC for Midterm

What you need to know for the Ethics Quiz

TL 5 Lecture

8 Areas of Ethical Responsibility

- Public
- Client and Employer
- Product
- Judgement
- Management
- Profession
- Colleagues
- Self

4 Factors in Measuring Copyright Fair Use

- The purpose of your use
- The nature of the copyrighted work
- The amount and substantiality of the portion taken
- The effect of the use upon the potential market

4 Areas of project risks

- Customer Mandate
- Scope and Requirements
- Execution
- Environment

Ethical use of AI in CS 3383

Using AI in class

U of I code of conduct: “Engaging in any behavior related to course completion prohibited by the instructor or otherwise including but not limited to unauthorized collaboration and reliance on prohibited technological assistance/artificial intelligence tools.”

Using AI in class

The use of AI in class is determined by your instructors.

Next semester there will be an AI infringement requirement
BC will chose one of the following:

- AI use prohibited
- AI with prior permission
- AI with Acknowledgement
- AI with No Acknowledgement

Using AI for your video game

As for using AI to help you make your game:

- Don’t use it for getting the job done

- If you do use it, make sure you’re learning something from what it tells you

- AI is an excellent learning tool

AI in videogames

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What is AI in Videogames?

Illusion of Intelligence

Believable environments and characters:

- Decision making
- Interaction and Responsiveness
- Navigation

NPC: Non-Player Character

AI Implementations:

Traditional AI

Phase 1

Scripted: Predefined rules

Pre-programmed conditions

Examples:

- Range Detection
- Fixed path
- Attack patterns

→ Good for simple mechanics
→ Can be limited and lacks realism

```cpp
#include <iostream>
using namespace std;

int main() {
    bool playerInRange = true;
    int enemyHealth = 100;

    if (enemyHealth <= 0) {
        cout << "Enemy is dead\n";
    }
    else if (playerInRange) {
        cout << "Enemy attacks player\n";
    }
    else {
        cout << "Enemy is patrolling\n";
    }

    return 0;
}
```

AI Implementations:

Systems-Based

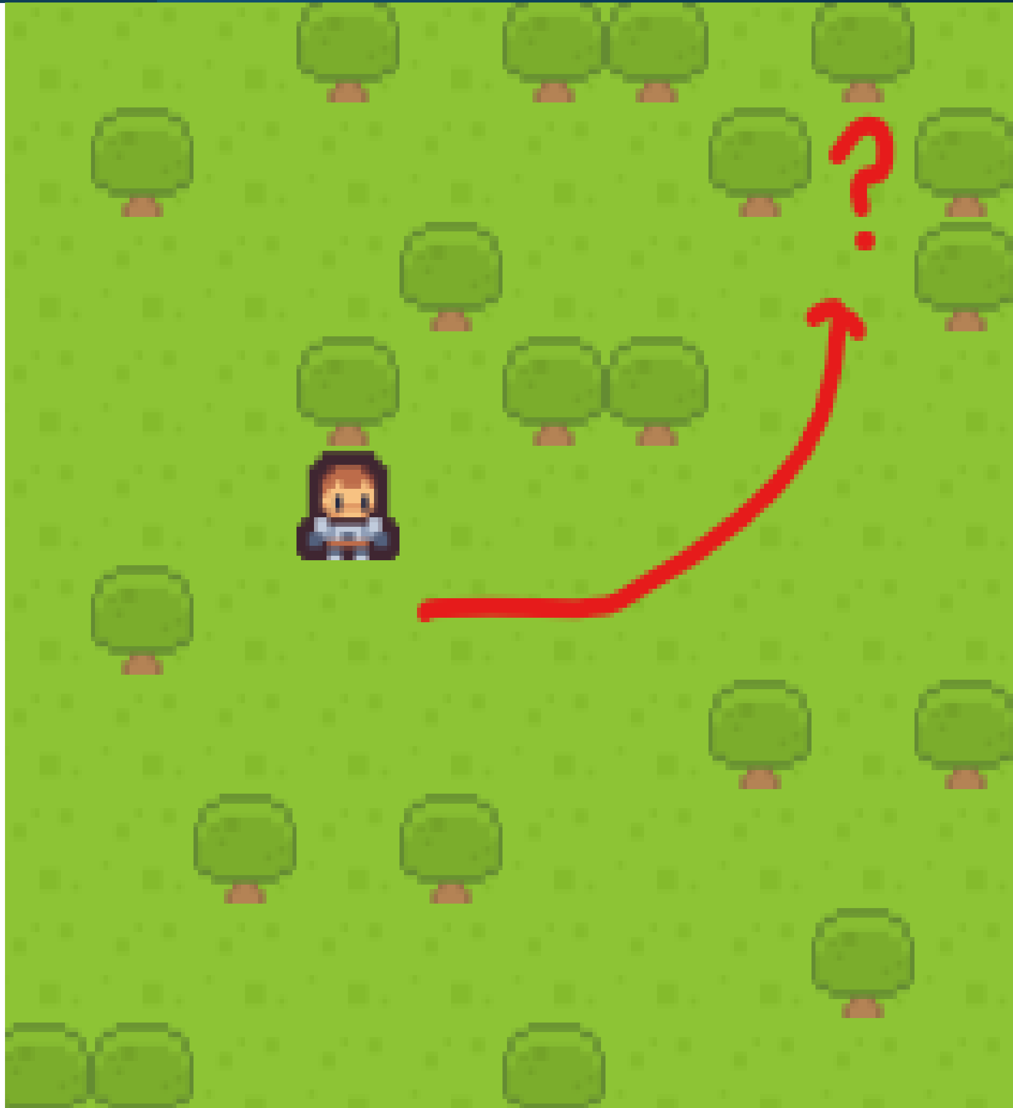
Phase 2

Behavior choice on multiple conditions

Techniques:

- Finite State Machines
  Patrol, Chase, Attack state switch
- Decision Trees
  Does x meet condition n?
  Each decision → different branch
- Pathfinding
  Dynamic Path calculation

→ More flexible, less predictable
→ Cannot adapt beyond predefinitions



AI Implementations:

Modern Day AI

Phase 3

Multimodal: Different types of data

Examples:

- Adaptive
  - Difficulty adjustment
  - Learning patterns

- Generative
  - Real Time responsiveness
  - Procedural elements

→ Closest to realism, little limitations
→ Risk of inappropriate behavior, requires safeguards and monitoring
→ Hardest to test and control

Tying AI to the 8 principles

- **Public:** Player experience with AI
- **Client and Employer:** Meet ethical goals in our AI
- **Product:** AI Testing
- **Judgement:** AI features
- **Management:** Ethical standards
- **Profession:** Don’t misuse AI
- **Colleagues:** Transparent when using LLMs
- **Self:** Adaptation with AI standards

Quiz Question:

The Quiz Question will include a statement describing an AI implementation scenario.

Example:

A game introduces an AI companion that adapts to players and generates responses, with advanced features only available with paid access.

You will have to respond the following questions related to it:

1. What kind of AI implementation is it?
2. Does it abide to the 8 ethical principles? Why or why not?

Dynamic Binding

```
using System;

public class MenuBase
{
    public virtual void HandlePrimaryAction()
    {
        Console.WriteLine("Default action");
    }
}

public class MainMenu : MenuBase
{
    public override void HandlePrimaryAction()
    {
        Console.WriteLine("Starting game");
    }
}

public class PauseMenu : MenuBase
{
    public override void HandlePrimaryAction()
    {
        Console.WriteLine("Resuming game");
    }
}
```

```
MenuBase currentMenu = new PauseMenu();
currentMenu.HandlePrimaryAction();
```

```
MenuBase currentMenu = new MainMenu();
currentMenu.HandlePrimaryAction();
```

Dynamic Binding

```csharp
using System;

public class MenuBase
{
    public void HandlePrimaryAction()
    {
        Console.WriteLine("Default action");
    }
}

public class MainMenu : MenuBase
{
    public void HandlePrimaryAction()
    {
        Console.WriteLine("Starting game");
    }
}

public class PauseMenu : MenuBase
{
    public void HandlePrimaryAction()
    {
        Console.WriteLine("Resuming game");
    }
}
```

```csharp
MenuBase currentMenu = new PauseMenu();
currentMenu.HandlePrimaryAction();
```

```csharp
MenuBase currentMenu = new MainMenu();
currentMenu.HandlePrimaryAction();
```

Dynamic Binding

```
public class Enemy
{
    4 references
    public virtual void Attack() { }
}

1 reference
public class Zombie : Enemy
{
    3 references
    public override void Attack() { }
}

1 reference
public class Archer : Enemy
{
    3 references
    public override void Attack() { }
}
```

```
Enemy enemy = new Zombie();
enemy.Attack();

enemy = new Archer();
enemy.Attack();
```

Thank you

Any questions?

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